

# SHABAA ALAM

Aeronautical Engineering Student | Imperial College London

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## EDUCATION

Imperial College London

Oct 2024 – Present (Exp. May 2028)

MEng Aeronautical Engineering (Year 3 of 4)

London, UK

- Seeking a 6 to 12-month engineering placement / Year in Industry role starting Summer 2027.
- Core Modules: Aerodynamics, Flight Dynamics & Control, Structural Analysis, Control Systems, CFD, FEA, Advanced Manufacturing.

## ENGINEERING & TECHNICAL PROJECTS

ICLR Ember Rocket Team (European Rocketry Challenge - EuRoC)

2025 – Present

Design & Manufacturing Engineer

- Designed 3D CAD assemblies for an automated carbon fibre winder in Fusion 360, selecting motor drive components and sizing belt-pulley power transmission systems for manufacturing feasibility.
- Machined stainless steel drive shafts on manual lathes within  $\pm 5 \mu\text{m}$  bearing tolerances to eliminate rotational vibration.

Duocopter Flight Controller System (2nd Year Final Project)

2026

Control Systems Engineer

- Modeled and simulated a discrete PID controller architecture in MATLAB & Simulink to stabilize a dynamic 43g dual-rotor UAV in hover.
- Optimized control parameters to minimize actuator energy (10.38 kJ over 70s), placing 2nd overall in group performance benchmarking.

Wind Turbine Rotor & Structural Analysis (1st Year Final Project)

2025

Structural & FEA Lead

- Analyzed blade structural integrity in Abaqus FEA, validating aerodynamic loading distributions against MATLAB analytical models.
- Investigated wind-tunnel test failure, tracing premature fracture to an inverted airfoil manufacturing error that exceeded predicted safety-factor limits at peak stress points.

Imperial Formula Racing (Formula Student)

2024 – 2025

Composites Engineer

- Engineered an ergonomic carbon fibre driver seat inlay using fire-retardant expanding foam tooling to comply with Formula Student fire safety regulations.
- Executed structural stress checks around seat mounting points against driver load constraints prior to composite layup and autoclave curing.

## HONORS & AWARDS

- **Arkwright Engineering Scholarship:** Awarded national scholarship providing industry mentorship and technical support.
- **Project of the Year Award - Leonardo (2023):** Won 1st place in EDT Industrial Cadets for leading mechanical design and 8 rapid-prototyping iterations of a 3D-printed rover suspension and wheel hub assembly.

## WORK EXPERIENCE & LEADERSHIP

Private & Academy Tutor (A-Level STEM) – Westcliff Academy for Learning

2023 – Present

- Tutor 10+ students weekly in A-Level Maths, Further Maths, and Physics, communicating complex technical concepts clearly.

Teaching Intern (STEM) – Westcliff High School for Boys

Jun 2025 – Jul 2025

- Mentored secondary students in analytical problem-solving and assisted in executing practical STEM workshops.

## TECHNICAL SKILLS & COMPETENCIES

**Coding & Software:** C++ (Numerical Methods), Python, MATLAB, Simulink

**Analysis & FEA/CFD:** Abaqus (FEA), STAR-CCM+ (CFD), Structural Analysis, Control Systems Engineering

**CAD & Design:** SolidWorks, Autodesk Fusion 360, Dassault 3DEXPERIENCE

**Manufacturing:** Manual Lathes & Milling, Carbon Fibre Composites, 3D Printing (FDM/ABS), Rapid Prototyping

**Languages:** English (Native), Bengali (Fluent)